A Product Counterexample to Arbitrarily Slow Proper L^p-Cocycles

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Status

Status: **counterexample_likely_valid**.

This packet concerns Question 1 in Section 6 of Antonio Lopez Neumann and Juan Paucar, *On growth of cocycles of isometric representations on L^p-spaces*, arXiv:2501.12808v2.

Source Question

The source paper defines a compactly generated locally compact group G to have arbitrarily slow proper cocycles in L^p if, for every proper function $f : G \to [0, \infty)$, there are an L^p-space E, a continuous isometric representation $\pi : G \to \mathcal{O}(E)$, and a proper cocycle $b \in Z^1(G, \pi)$ such that $\|b(g)\| \leq f(g)$ for every $g \in G$.

Question 1 asks:

> Let G be a locally compact compactly generated group. Suppose that G is *a-FL^p-menable* (i.e. G admits a proper cocycle on an L^p-space) for some $2 < p < \infty$, not *a-T-menable* and does not have property (T). Does G have arbitrarily slow proper cocycles in L^p?

The question appears on page 28 of the locally rendered source PDF:

5.3 Large equivariant compression

We will now prove Theorem E.

Proof of Theorem E. Let G be a compactly generated locally compact group and suppose that there exist a continuous isometric representation $\pi : G \to \mathcal{O}(E)$ on a q-uniformly smooth Banach space E and a 1-cocycle $b \in Z^1(G, \pi)$ such that:

$$\rho_b(n) = \inf_{|g|_S = n} ||b(g)||_E \succ n^{1/q}.$$

The contrapositive of Proposition 5.4 allows one to conclude that $l(\mu) = 0$. If G is a discrete group, it is classical to show that $l(\mu) = 0$ implies that (G, μ) has the Liouville property. For example, this can be shown as a combination of [KV83, 1.1] and the computation in [KL07, Section 4].

If G is a compactly generated locally compact group, one can define the asymptotic entropy $h(\mu)$ of the random walk driven by μ by using densities as in [Der06, Section III]. One can reproduce the argument of [KL07, Section 4] in the locally compact case to show that the condition $l(\mu) = 0$ still implies that $h(\mu) = 0$. We obtain that (G, μ) has the Liouville property using [Der06, Théorème p.255]. $\square$

6 Questions

As in Proposition 4.7, we say that a locally compact compactly generated group G *has arbitrarily slow proper cocycles in* L^p (where $1 < p < \infty$) if for every proper function $f : G \to [0, \infty)$, there exist some L^p-space E, a continuous isometric representation $\pi : G \to \mathcal{O}(E)$ and a proper cocycle $b \in Z^1(G, \pi)$ such that $||b(g)|| \leq f(g)$.

We know that a-T-menable groups have arbitrarily slow proper cocycles in L^p for every $1 < p < \infty$ [MdlS23, Proposition 3.1]. We also know that property (T) groups do not have arbitrarily slow proper cocycles in L^p for every $1 < p < \infty$ because of Theorem D.

Question 1. *Let G be a locally compact compactly generated group. Suppose that G is a-FL^p-menable (i.e. G admits a proper cocycle on an L^p-space) for some $2 < p < \infty$, not a-T-menable and does not have property (T). Does G have arbitrarily slow proper cocycles in L^p?*

Examples of groups with conditions as in Question 1 include products of a-T-menable groups with property (T) groups, such as $\mathrm{SO}(n, 1) \times \mathrm{Sp}(m, 1)$ for $n, m \geq 2$.

Some of the techniques in Section 4 work not only for L^p-spaces, but for the larger class of p-uniformly convex Banach spaces. An important ingredient in the proof of Theorem D is that every cohomology class of all isometric representations on L^p-spaces contain harmonic cocycles because the group in question has property (T).

Question 2. *Let G be a locally compact second countable group with property (T). Do all continuous isometric representations of G on p-uniformly convex spaces have a harmonic 1-cocycle in every first cohomology class?*

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Proof Intuition

The obstruction is already present in the product examples mentioned by the source paper. Put a property (T) factor B next to a non-(T) a-FL^p-menable factor A. The product $A \times B$ is still a-FL^p-menable and is not a-T-menable, but any proper cocycle on the product must be unbounded on the B-axis. Now choose the demanded "arbitrarily slow" majorant to grow only logarithmically in that B direction. The source paper's property-(T) theorem forces every unbounded L^p-cocycle of B to have random-walk L^p-average at least $n^{1/p}$, contradicting the logarithmic upper bound.

Counterexample

Theorem 1. *Fix $2 < p < \infty$. Let B be a noncompact compactly generated locally compact group with property (T) that is a-FL^p-menable. Let A be a noncompact compactly generated locally compact group without property (T) that is a-FL^p-menable. Then $G = A \times B$ satisfies the hypotheses of Question 1 for this p, but G does not have arbitrarily slow proper cocycles in L^p.*

Proof. The product $G = A \times B$ is compactly generated. Since both factors admit proper L^p-cocycles, their ℓ_p-direct sum is a proper L^p-cocycle on G, so G is *a*-FL^p-menable. The quotient map $G \to A$ shows that G does not have property (T). Also G is not *a*-T-menable: otherwise the closed subgroup B would be *a*-T-menable; combined with property (T), this would force B to be compact, contradicting the hypothesis.

Choose compact generating sets S_A, S_B and let $|\cdot|_A, |\cdot|_B$ be the associated word lengths. Define

$$f(a, b) = 1 + |a|_A + \log(1 + |b|_B), \qquad (a, b) \in A \times B.$$

This is a proper function on G, because its sublevel sets are contained in products of word balls in A and B.

Assume, toward a contradiction, that G has arbitrarily slow proper L^p-cocycles. Then there are an L^p-space E, a continuous isometric representation $\pi : G \to \mathcal{O}(E)$, and a proper cocycle $c \in Z^1(G, \pi)$ such that

$$\|c(a, b)\| \leq f(a, b) \qquad \text{for all } (a, b) \in G.$$

Restrict to the B-factor:

$$c_B(b) = c(e_A, b), \qquad b \in B.$$

This is a 1-cocycle for the restricted representation of B. It is proper, hence unbounded, because the inclusion $B \to A \times B$, $b \mapsto (e_A, b)$, is proper and c is proper. In particular c_B is not a coboundary, since every coboundary $b \mapsto v - \pi(b)v$ is bounded.

By the separable reduction in Appendix A of the source paper, the restricted cocycle may be regarded inside a separable L^p-subspace invariant under B. Since B has property (T) and is *a*-FL^p-menable, it does not have property FL^p. The source paper's Theorem D therefore applies to the unbounded cocycle c_B: for every compactly supported cohomologically adapted probability measure μ on B, there is a constant $C > 0$ such that

$$\|c_B\|_{L^p(\mu^{*n})} \geq C^{-1} n^{1/p} \qquad \text{for all } n \geq 1.$$

Choose such a measure μ supported in a compact generating set contained in S_B. Then μ^{*n} is supported in the word ball $\{b \in B : |b|_B \leq n\}$. The assumed pointwise bound gives

$$\|c_B(b)\| \leq 1 + \log(1 + |b|_B) \leq 1 + \log(1 + n) \qquad (b \in \operatorname{supp} \mu^{*n}),$$

and hence

$$\|c_B\|_{L^p(\mu^{*n})} \leq 1 + \log(1 + n).$$

This contradicts $C^{-1} n^{1/p} \leq 1 + \log(1+n)$ for large n. Thus the proper function f is not dominated by any proper L^p-cocycle on G, and G does not have arbitrarily slow proper cocycles in L^p. $\square$

Corollary 1. *For suitable sufficiently large p, the product groups* $\mathrm{SO}(n, 1) \times \mathrm{Sp}(m, 1)$*, $n, m \geq 2$, are negative answers to Question 1.*

Proof. The source paper itself lists these products as examples satisfying the hypotheses of Question 1. More explicitly, $\mathrm{SO}(n,1)$ is a-T-menable and does not have property (T), while $\mathrm{Sp}(m,1)$ is noncompact and has property (T). For sufficiently large p, the rank-one hyperbolic factor $\mathrm{Sp}(m,1)$ admits a proper affine isometric action on an L^p-space by the standard Bourdon–Pajot/Pansu L^p-cohomology construction cited in the source paper. The theorem applies. □

Verification Notes

The argument uses no computation. The only growth comparison is the elementary asymptotic contradiction $n^{1/p} \neq O(\log n)$.

Duplicate and novelty checks were bounded as follows. I searched the cheap run indexes for arXiv:2501.12808, "arbitrarily slow proper cocycles", "slow proper cocycles", "a-FLp̂", and the product examples; no prior packet or exact duplicate was found. Web searches on 2026-07-03 for the exact question and close variants found the source paper, arXiv:2501.12808, but no later exact resolution. The novelty confidence is therefore *likely valid / needs human review*, not human-verified.

Human Review Recommendation

Check two points first: (1) that the source paper's Theorem D is being applied with the intended separable reduction to the restricted B-cocycle, and (2) that the definition of "arbitrarily slow proper cocycles in L^p" allows arbitrary proper control functions such as $1 + |a|_A + \log(1 + |b|_B)$. If both are accepted, the contradiction is direct.

References

[1] Antonio Lopez Neumann and Juan Paucar. *On growth of cocycles of isometric representations on L^p-spaces.* arXiv:2501.12808v2, 2026.

[2] Uri Bader, Alex Furman, Tsachik Gelander, and Nicolas Monod. Property (T) and rigidity for actions on Banach spaces. *Acta Mathematica* 198 (2007), 57–105.

[3] Marc Bourdon and Herve Pajot. Cohomologie l_p et espaces de Besov. *Journal fur die reine und angewandte Mathematik* 558 (2003), 85–108.

[4] Bachir Bekka, Pierre de la Harpe, and Alain Valette. *Kazhdan's Property (T).* Cambridge University Press, 2008.